Require: request r; the store S
Ensure: one uniform reply, after a send, a downgrade, a delay or a refusal

procedure ProcessRequest(r)
  $p \leftarrow$ Attest(r); **if** $p = \bot$ **or** $r.\text{address} \in \mathcal{V}$ **then return** Refuse()
  if $\neg$ValidSession(r) **or** $n_{\text{sess}}[r.\text{session}] \geq 3$ **then return** Refuse()
  if Denylisted(r) **or** $\neg$Take($r.\text{address}, r./24, r.\text{ASN}$) **then return** Refuse()
  if $p = \text{web}$ **and** $(\text{score}(r) < \theta_w$ **or** $\text{origin}(r) \notin \mathcal{O})$ **then return** Refuse()
  $(c, \text{live}, v) \leftarrow$ CheckNumber($r.\text{number}$); **if** $\neg\text{live}$ **or** $c \in \{\text{off-list}, \text{premium}\}$ **then return** Refuse()
  if $\neg$ValidBody(r) **or** $r.\text{number} \in \mathcal{X}$ **or** $((r.\text{bulk} \vee r.\text{test}) \wedge \neg r.\text{credential})$ **then return** Refuse()
  $s \leftarrow$ Equation (1); $t \leftarrow$ Tier($s, v, r.\text{history}$)
  if $t = \text{block}$ **then return** Refuse(); **if** $t = \text{challenge}$ **then return** Challenge()
  if $\neg$Take(backoff$[r.\text{number}]$) **then return** Refuse()
  if $\neg$Take($m \cdot K[r.\text{source}, p, \text{country}(r.\text{number})]$) **then return** Refuse()
  mode $\leftarrow$ Mode(budget); **if** mode $=$ emergency **then return** Refuse()
  $(\text{ok}, g) \leftarrow$ Reserve(count, spend, slot$[r.\text{block}]$); **if** $\neg\text{ok}$ **then return** Refuse()
  $t \leftarrow$ Raise($t, g, r.\text{history}$); **if** $t = \text{challenge}$ **then return** Challenge()
  Dispatch(r, t); Write(S)
  return Reply()

procedure OnOutcome(x, o)
  for all $k \in \text{keys}(x)$ **do** $n_k \leftarrow n_k + 1$; $v_k \leftarrow v_k + [o = \text{verified}]$; **if** $n_k \geq 50$ **and** $v_k < 0.1\, n_k$ **then** Denylist$(k, 24\,\text{h})$
  if $o = \text{verified}$ **then** $\text{trusted}[x.\text{number}] \leftarrow \text{true}$
  $b \leftarrow x.\text{block}$; $S_c[b] \leftarrow \max(-ch,\ S_c[b] + \delta_c(o))$; **if** $o = \text{verified}$ **then** $S_s[b] \leftarrow \max(-ch,\ S_s[b] + \delta_s(x))$
  if $S_c[b] > h$ **or** $S_s[b] > h$ **then** Verdict$(b, 1\,\text{h})$; $S_c[b], S_s[b] \leftarrow 0$